



Effects of planting date and variety on flooded rice production in the deepwater area of Thailand

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ABSTRACT

Crop management plays an important role in the transition from a deepwater rice to a flooded rice production system but information about optimum management strategies are currently lacking. The goal of this study was to determine the effect of planting date and variety on flooded rice production in the deepwater area of Thailand. Two experiments were conducted at the Bang Taen His Majesty Private Development Project in 2009 and 2010 to represent conditions prior to flooding (early rainy season) and after flooding (dry season). The early rainy season crop covered the period from May to October 2009, while the dry season crop covered the period from November 2009 to April 2010. The experimental design was a split plot with four main plots and three sub plots replicated four times. The treatments for the main plot were various planting dates, while the treatments for the sub plots were rice varieties. The dates of the critical developmental phases of rice were recorded and biomass was sampled during the growing period. The collected data were statistically analyzed using ANOVA and treatment means were compared to identify the appropriate planting date and the best variety for the area. The highest average yield was obtained for variety PSL2 across transplanting dates from June 19 to July 23, with an average yield of 3898 kg ha⁻¹. The dry season crop showed that both biomass and yield were affected by the interaction between planting date and variety. The highest yield was obtained for variety PTT1 transplanted on November 9. The research showed that the variety PSL2 is the most suitable variety for early rainy season production with a transplanting date ranging from June 19 to July 23, while the variety PTT1 planted on November 9 was the best management practice for the dry season crop. However, a high yielding flooded rice variety that has a short growth duration is still needed for this area.

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1. Introduction

In the eastern plain of Thailand, deepwater rice production (DWR) is the main management practice used during the rainy season (Catling, 1992). A range of photo-sensitive rice varieties with appropriate harvesting dates are planted in the lowest terrace from April to May (Department of Agriculture, 2004a,b). Flooding begins in August and continues until the maximum flooding depth, at least one meter, is reached in November. The rice is harvested from late November to January (Kupkanchanakul et al., 1986; Puckridge et al., 1994). However, potential yield of deepwater rice is relatively low compared to other rice production systems. Therefore, many deepwater rice farmers in Thailand, Vietnam, and other region with

similar condition are converting their fields to flooded rice production (FDR) to increase potential yield, resulting in a higher economic return (Catling, 1992; Mahabub et al., 1994; Puckridge et al., 1989).

A flooded rice plant is a dwarf erect plant, 80–120 cm height, that is not photoperiod sensitive and can be grown throughout the year. Depending on the variety, growth duration varies from 90 to 140 days (Department of Agriculture, 2002). The significant FDR characteristic is a high yield compared to DWR. However, the constraints of flooded rice production in a deepwater area are the onset and receding of flood water during the rainy season and water supply during the dry season.

Differences in planting dates are affected by many factors that include temperature, solar radiation, rainfall, and soil condition (Schafera and Kirchoff, 2000; Shunji and Kimuraa, 2007). Late planting can delay panicle initiation, heading, and maturity (Halder et al., 2004). The objective of this study was to determine the effect of variety used, and planting date on growth and yield of flooded rice production in a deepwater area. This analysis can then be used to provide farmers from areas ranging from Southeast Asia to Africa

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Table 1

Comparison of harvest dates for three rice varieties grown in the deepwater area of Prachin Buri, Thailand during 2009 and 2010.

Planting date	Seeding date	Transplanting date	Harvesting date		
			CNT1	PTT1	PSL2
Early rainy season					
PD1	May 20, 2009	June 19, 2009	September 23, 2009	September 21, 2009	September 17, 2009
PD2	June 3, 2009	July 2, 2009	October 2, 2009	September 30, 2009	September 26, 2009
PD3	June 24, 2009	July 16, 2009	October 14, 2009	October 21, 2009	October 14, 2009
PD4	July 1, 2009	July 23, 2009	October 24, 2009	October 26, 2009	October 21, 2009
Dry season					
PD1	November 9, 2009	December 9, 2009	February 26, 2010	March 8, 2010	February 26, 2010
PD2	November 23, 2009	December 21, 2009	March 8, 2010	March 21, 2010	March 8, 2010
PD3	December 8, 2009	January 7, 2010	March 24, 2010	April 7, 2010	March 24, 2010
PD4	December 21, 2009	January 18, 2010	Apr 15, 2010	April 19, 2010	April 15, 2010

with appropriate information so that they can make the best choices when changing from deepwater rice to a flooded rice production system.

2. Materials and methods

2.1. Field experiments

Field experiments were conducted at the Bang Taen His Majesty Private Development Project, Bansang, Prachin Buri, Thailand, located at a latitude of 13° 52' N, a longitude of 101° 09' E and an elevation of two meters above mean sea level. The experimental design was a split plot in a randomized complete block with four replications. Planting date (PD) was the main plot and the variety (V) was the sub-plot and was randomized within each planting date. There were a total of four planting dates and three rice varieties. The experiment was conducted during two seasons, i.e., the early rainy season (ERS) and dry season (DS). For both cropping seasons, the same three rice varieties were grown which consisted of Chai Nat 1 (CNT1), Pathum Thani 1 (PTT1) and Pitsanulok 2 (PSL2). For the early rainy season experiment, the first planting date (PD1) was started on May 20 followed by PD2 on June 3, PD3 on June 24, and PD4 on July 1, 2009. The crop for the first planting date was harvested on September 17 and the crop for the last planting date was harvested on October 21, 2009. The dry season crop started in November. The depth of flood was declined. Water level in the field was suitable for land preparation and planting. The first crop PD1 was planted on November 9 followed by November 23, December 8 and December 21, 2009 for PD2, PD3 and PD4, respectively. The first harvest for the dry season crop started on February 26, 2010 and ended on April 15, 2010 for PD4 crop (Table 1). For water management, there were two sources of water that were used for flooded rice production: water from natural rainfall and water pumped from the Bang Pakong River. Both sources were used for the early rainy season crop. For the dry season crop, the only source of water was pumped from the river. Therefore, both the quantity and the quality (the level of salinity) of water were critical factor for dry season production.

Composite soil samples were collected from different soil depths prior to plowing the field (Hoogenboom et al., 1999). Land preparation was conducted with a seven disk implement for the first plowing and a hand rototiller for the second plowing within one week after the first plowing, followed by harrowing. After land preparation was completed, the small rice plants were transplanted at a spacing of 20 cm × 20 cm with three seedlings per hill. The area of the main plot was 7.0 m × 8.8 m, with three sub plots measuring 7.0 m × 2.8 m. The border between each main plot was one meter. Each sub plot contained 15 rows with 36 hills per row. Continuous flooding was maintained at the depth of 10 cm. There were two applications of chemical fertilizer; the first one was applied at a rate of 30 kg N and 36 kg P₂O₅ ha⁻¹ one week after transplanting and the second one was applied at a rate of 29 kg N ha⁻¹ at panicle initiation (PI) (Department of Agriculture, 2004a,b).

2.2. Data collection

A rice plant has three major developmental phases: the vegetative phase, the reproductive phase and the ripening phase (Department of Agriculture, 2004a,b; Kunnut, 2006). The vegetative phase covers the period from germination to panicle initiation. The reproductive phase covers the period from panicle initiation to flowering phase and the ripening phase covers the period from flowering to maturity. The developmental stages and associated dates were monitored, including the panicle initiation date, flowering date, the milky date and the maturity or harvesting date. The panicle initiation date was monitored daily starting at 55 days after planting until appearance of the panicle primordium. For the other developmental stages, a five-day interval was used for monitoring. Biomass samples for growth analysis were collected throughout the growing season (Hoogenboom et al., 1999). Biomass samples were collected five times, including at the seedling stage (a stage during the vegetative phase), panicle initiation stage, flowering stage, milky stage and maturing stage (Department of Agriculture, 2004a,b). One hundred plants for each variety were collected at the seedling stage before transplanting. For the panicle initiation stage, flowering stage and milky stage, 15 hills per plot were sampled. At

Table 2

Soil analysis of experimental plot in the deepwater area of Prachin Buri, Thailand sampled in 2009.

Soil depth (cm)	pH	OM (%)	N (%)	P (ppm)	K (ppm)	Soil texture
0–15	4.0	2.3	0.1	9.6	81.3	Sandy clay loam
15–30	3.8	1.0	0.1	4.5	86.3	Sandy clay loam
30–45	3.2	0.4	0.0	2.3	78.7	Sandy clay loam
45–60	3.2	0.3	0.0	2.5	85.0	Sandy clay loam
Average	3.6	1.0	0.1	4.7	82.8	
SD	0.4	0.9	0.05	3.4	3.5	

Note: pH, 1:1 (H₂O); N, total N; P, available P; and K, extractable K.

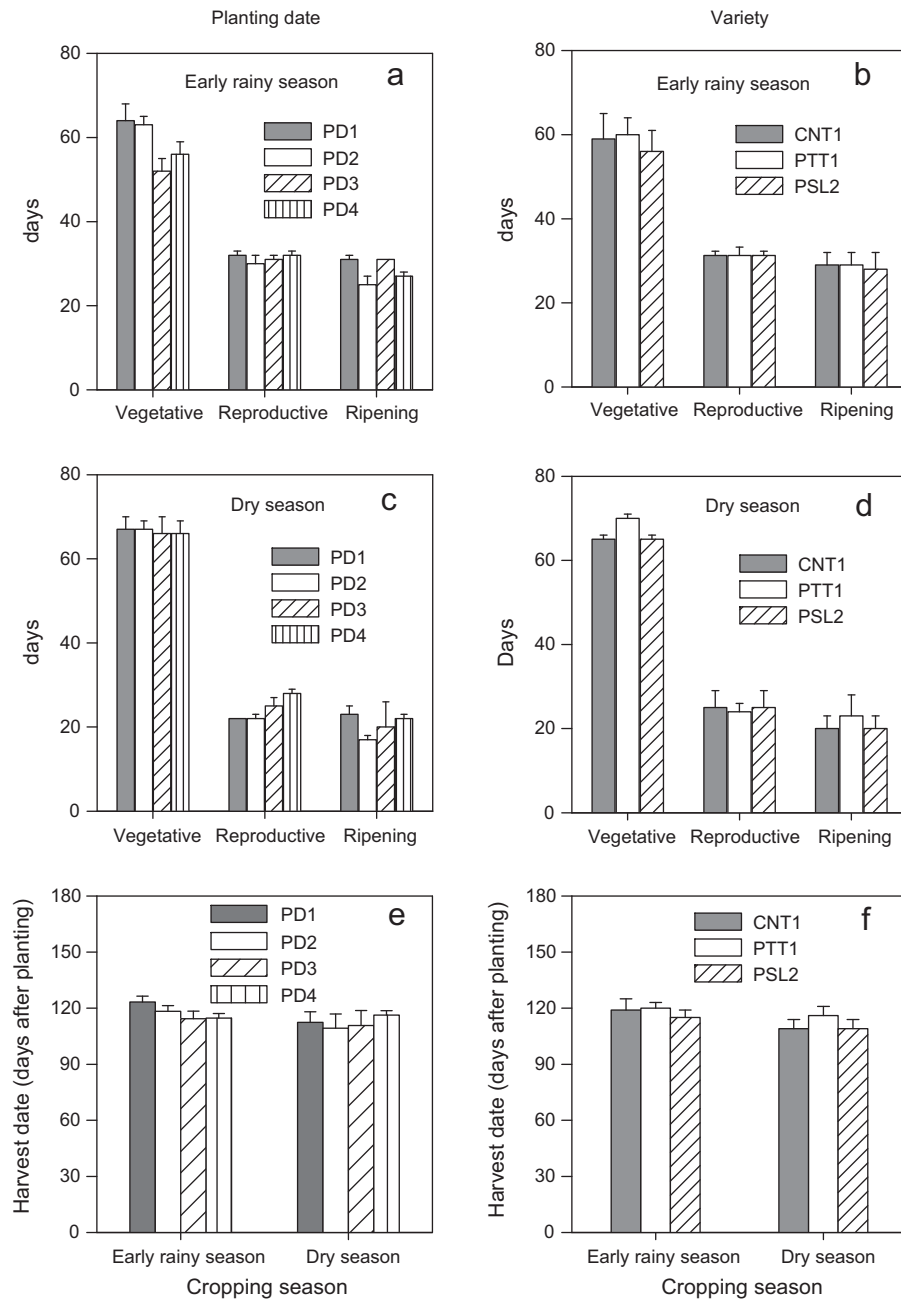


Fig. 1. Development phase and growth duration for four planting dates and three varieties for the early rainy season (ERS) and dry season (DS) crop, average growth duration under four planting dates for ERS (a) and DS (c); average growth duration for three varieties for ERS (b) and DS (d), average harvest date under four planting dates (e) and three varieties dates (f).

final harvest, 25 hills were sampled for biomass, yield and yield component evaluation. The individual stem, leaf, and panicle components of the sampled rice hills were separated and dried in a hot air oven at 75 °C for 72 h until constant weight and then weighed for further growth analysis.

2.3. Data analysis

The experimental data were analyzed using statistical analysis techniques including arithmetic mean, standard deviation and analysis of variance (ANOVA) for treatment means comparison.

The arithmetic mean was defined as per Miller et al. (2009)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

The standard deviation (SD) was defined as per Gravetter and Wallnau (2008)

$$SD (\sigma) = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}} \quad (2)$$

The ANOVA technique was used to compare the effect of planting date and variety on growth and yield of flooded rice production in deepwater area. The linear effects model for a split-plot design is shown below (Gomez and Gomez, 1984; William and Gettinby, 1998):

$$x_{ijk} = \mu + \rho_i + \alpha_j + \rho\alpha_{ij} + \beta_k + \alpha\beta_{jk} + \rho\alpha\beta_{ijk} \quad (3)$$

where $i = 1, 2, \dots, n$, $j = 1, 2, \dots, a$, and $k = 1, 2, \dots, b$,

Table 3
Analysis of variance (P-value) for biomass, yield and harvest index (HI) for the early rainy season and dry season crop in the deepwater area of Prachin Buri, Thailand during the 2009 and 2010 growing seasons.

Treatment factor	Leaf biomass				Stem biomass				Above ground biomass				Yield	HI
	S1	S2	S3	S4	S1	S2	S3	S4	S1	S2	S3	S4		
Early rainy season crop														
Planting date (PD)	0.0000	0.1117	0.0377	0.0713	0.000	0.1699	0.0262	0.3321	0.0000	0.2553	0.0306	0.4465	0.0942	0.1670
Variety (V)	0.2450	0.0000	0.0331	0.0000	0.6292	0.0389	0.3630	0.0000	0.9672	0.6731	0.0338	0.0000	0.0055	0.0000
PD × V	0.1738	0.0003	0.0004	0.0088	0.6890	0.0370	0.3957	0.0190	0.4427	0.3574	0.2347	0.0103	0.2214	0.2618
%CV	11.11	10.55	13.74	8.15	11.94	13.69	16.86	13.86	12.39	12.42	14.66	5.46	7.84	8.82
Dry season crop														
Planting date (PD)	0.0049	0.1711	0.0511	0.1089	0.0000	0.0005	0.0007	0.0000	0.5001	0.0038	0.0004	0.0000	0.0000	0.0000
Variety (V)	0.2946	0.0857	0.0020	0.1492	0.2719	0.2058	0.4027	0.1217	0.2988	0.4440	0.6985	0.4408	0.0070	0.0000
PD × V	0.6611	0.2521	0.0001	0.1287	0.3806	0.0248	0.9178	0.1581	0.4358	0.2031	0.8495	0.0063	0.0031	0.0004
%CV	17.48	14.99	9.78	11.16	17.30	17.61	21.61	15.40	16.01	15.30	17.55	9.81	14.57	11.17

Note: S1, panicle initiation stage; S2, flowering stage; S3, milky stage; and S4, harvesting stage.

The above linear model can be explained with the following equation (Williams et al., 2002):

$$\begin{aligned} \text{Observation} = & (\text{overall mean}) + (\text{replication effect}) + (\text{A effect}) \\ & + (\text{main-plot residue}) + (\text{B effect}) \\ & + (\text{interaction between A and B effect}) \\ & + (\text{sub-plot residue}). \end{aligned}$$

3. Results

3.1. Soil properties

The field experiments were conducted on an acid sulfate soil that is part of the Rang sit (Rs) soil series. The Rs soil series is under the Vertisols soil order, sub order Aquerts, great group of Dystraquerts, and Sulfaquepts of sub group soil. The Rs soil series is in the family of Fine, Mixed, Isohyperthermic (Vijarnsorn and Eswaran, 2002). The soil texture is a sandy clay loam with average ($\pm$ SD) pH (1:1 H₂O) at a soil depth of 0–60 cm of 3.6 ($\pm$ 0.40), soil organic matter of 1.0 ($\pm$ 0.90)%, total N of 0.1 ($\pm$ 0.05)%, available P of 4.7 ($\pm$ 3.4) ppm and extractable K of 82.8 ($\pm$ 3.5) ppm (Table 2). The pH of the top soil layer (0–15 cm) was 4.0, with a soil organic matter of 2.3% and available P of 9.6 ppm.

3.2. Effect of planting date and variety on development

3.2.1. Planting date

The average duration and SD of the vegetative phase for the ERS for the four planting dates over the three varieties was 64 ($\pm$ 4), 63 ($\pm$ 2), 52 ($\pm$ 3) and 56 ($\pm$ 3) days for PD1, PD2, PD3 and PD4, respectively. The average duration and SD of the reproductive phase for ERS for the four planting dates over the three varieties was 32 ($\pm$ 1), 30 ($\pm$ 2), 31 ($\pm$ 1) and 32 ($\pm$ 1) days for PD1, PD2, PD3 and PD4. The average duration and SD of the ripening phase for ERS was 31 ($\pm$ 1), 25 ($\pm$ 2), 31 ($\pm$ 0) and 27 ($\pm$ 1) days for PD1, PD2, PD3 and PD4, respectively (Fig. 1a). There was no significant difference of the planting date on the duration of the developmental phases of flooded rice production in the deepwater area for ERS. However, the vegetative phase tended to be shorter when the planting date was delayed (PD3 and PD4) and consequently affected the overall harvest date. The overall growth duration for PD1 was 123 ($\pm$ 3) days which was not significantly difference from PD2 and was longer than the growth duration for PD3 (114 $\pm$ 4) and PD4 (115 $\pm$ 3) (Fig. 1e).

The average duration and SD of the vegetative phase for the DS for the four planting dates over three varieties was 67 ($\pm$ 3), 67 ($\pm$ 2), 66 ($\pm$ 4) and 66 ($\pm$ 3) days after germination for PD1, PD2, PD3 and PD4, respectively. The average duration and SD of the reproductive phase for the four planting dates over three varieties was 22 ($\pm$ 0), 22 ($\pm$ 1), 25 ($\pm$ 2) and 28 ($\pm$ 1) days for PD1, PD2, PD3 and PD4, respectively. The average duration of ripening phase was 23 ($\pm$ 1), 17 ($\pm$ 2), 20 ($\pm$ 0) and 22 ($\pm$ 1) days of PD1, PD2, PD3 and PD4, respectively (Fig. 1c). The overall growth durations were not statistically different, but the harvest date for the DS tended to be shorter than for the ERS (Fig. 1e).

3.2.2. Variety

The average duration and SD of the vegetative phase for the ERS for the three varieties over four planting dates was 59 ($\pm$ 6), 60 ($\pm$ 4), and 56 ($\pm$ 5) days after germination for CNT1, PTT1, and PSL2, respectively (Fig. 1b). The average duration and SD of the reproductive phase for the ERS for the three varieties over four planting dates was 31 ($\pm$ 1), 31 ($\pm$ 2), and 31 ($\pm$ 1) days for CNT1, PTT1, and PSL2, respectively. The average duration of ripening phase was 29 ($\pm$ 3), 29 ($\pm$ 3), and 28 ($\pm$ 4) for CNT1, PTT1, and PSL2, respectively (Fig. 1b).

Table 4

Comparison of biomass and yield of flooded rice under difference planting dates in the deepwater area of Prachin Buri, Thailand during the 2009 and 2010 growing seasons.

Planting dates	Biomass ^a (kg ha ⁻¹)	Yield ^a (kg ha ⁻¹)
Early rainy season		
May 20	8703 (±332)a	3932 (±287)a
June 3	8197 (±24)a	3671 (±335)ab
June 24	8522 (±285)a	3565 (±167)b
July 1	8605 (±950)a	3746 (±92)ab
Average	8507 (±219)	3729 (±154)
P-value		
Dry season		
November 9	8284 (±533)a	3441 (±222)a
November 23	6894 (±84)b	2536 (±280)b
December 8	4117 (±792)c	465 (±390)c
December 21	2679 (±137)d	70 (±17)d
Average	5494 (±2554)	1628 (±1622)
P-value	0.0000	0.0000

^a Means with a column followed by the same letter are not significantly different according to LSD test at $P < 0.05$.

The average duration of the development phase from germination to harvest among the three flooded rice varieties and different planting dates was 118 (±5) days after planting. The average harvest date of the three varieties over four planting dates was 119 (±6), 120 (±3), and 115 (±4), of CNT1, PTT1, and PSL2, respectively (Fig. 1f). The average duration of the vegetative phase (germination to panicle initiation: PI), the reproductive phase (from PI to flowering stage: FD) and the ripening phase (from FD to harvesting: HD) (Department of Agriculture and Food and Agriculture Organization, 2004) were not significantly different among the development phases of the three varieties (Fig. 1b).

The average duration and SD of the vegetative phase for the DS for the three varieties dates over the four planting dates was 65 (±1), 70 (±1), and 65 (±1) days after germination for CNT1, PTT1, and PSL2, respectively (Fig. 1d). The average duration and SD of the reproductive phase for DS for the three varieties over the four planting dates was 25 (±4), 24 (±2), and 25 (±4), days for CNT1, PTT1, and PSL2, respectively. The average duration of ripening phase was 20 (±3), 23 (±5), and 20 (±3), respectively (Fig. 1d). The average harvest date among the three varieties for the four planting dates was 109 (±5), 116 (±5), and 109 (±5) days after planting for CNT1, PTT1, and PSL2, respectively (Fig. 1f). The overall average harvest date of the three varieties over four planting dates of flooded rice production after flooding was 111 (±6) days after planting.

3.3. Effect of planting date and variety on growth and biomass

3.3.1. Planting date

There was a significant effect ($P < 0.05$) of the planting date on leaf and stem biomass and consequently above ground biomass for the first growth analysis sampling at panicle initiation stage (S1) of the ERS crop. For the growth analysis samples taken at the flowering (S2) and harvest stage (S4), leaf biomass, stem and above ground biomass were not significantly different ($P > 0.05$; Table 3). Above ground biomass was very similar (Table 4) across planting dates of flooded rice from May 20 to July 1. At final harvest, the P value for above ground biomass was 0.4465 indicating that there was no significant difference among planting dates (Table 3).

For the growth analysis of the DS, stem biomass was significantly different for all four growth analysis samples. Leaf biomass was significantly different for the first sampling (S1) at the panicle initiation stage ($P < 0.05$) but there was no effect for the other three stages ($P > 0.05$). Total above ground biomass was significantly different ($P < 0.05$) for the growth analysis samples taken at the S2, S3,

and S4 development stages, while above ground biomass at the S1 stage was not significantly different ($P = 0.5001$) (Table 3).

3.3.2. Variety

Leaf biomass of flooded rice during the early rainy season was significantly different ($P < 0.05$) at the S2, S3 and S4 development stages but it was not significantly different at S1 ($P = 0.2450$). Stem biomass at S2 and S4 and above ground biomass at S3 and S4 were also significantly different ($P < 0.05$). At final harvest, leaf, stem and above ground biomass were all significantly different ($P < 0.05$; Table 3). This indicated that the three rice varieties were significantly different in term of growth for the early rainy season crop. The PTT1 variety produced the largest amount of above ground biomass, with an average of 9022 kg ha⁻¹ followed by CNT1 with 8509 kg ha⁻¹ and PSL2 with 8004 kg ha⁻¹, respectively (Table 5).

Leaf biomass of flooded rice during the DS at the S1, S2 and S4 development stages, and stem and above ground biomass at all stages were not significantly different ($P > 0.05$). Leaf biomass at the S3 development stage was only significantly different ($P = 0.0020$) (Table 3). Overall, there was no effect of rice varieties on above ground biomass of flooded rice production during the dry season for the deepwater area.

3.4. Effect of planting date and variety on final yield

3.4.1. Planting date

There was no effect of planting date on yield for the early rainy season crop. Average yield (at a 14% of moisture content) for the four plantings, i.e. 3932 kg ha for PD1, 3671 kg ha for PD2; 3565 and 3746 kg ha⁻¹ date was not significantly different ($P = 0.0942$). However, there was an effect of planting dates on flooded rice yield during the dry season crop (Fig. 2). Average yield for the four planting dates, i.e., 3441 kg ha⁻¹ for PD1; 2536 kg ha⁻¹ for PD2 was significantly different ($P = 0.0000$) (Table 4).

3.4.2. Variety

The variety that was planted affected yield for both the ERS and the DS crop with a P -value of 0.0055 and 0.0070, respectively. The variety PSL2 had the highest yield for the early rainy season crop over four planting dates with an average of 3898 kg ha⁻¹. However, it was not significantly different from the variety PTT1 that had a yield of 3786 kg ha⁻¹. The harvest index (HI) for the varieties PSL2 and PTT1 were 0.49 and 0.42, respectively. The variety CNT1 had the lowest yield with an average yield of 3529 kg ha⁻¹ and a harvest index of 0.42 (Table 5). For the dry season crop, the highest yield over four planting dates at a 14% of moisture content was obtained with the variety CNT1 with an average yield of 1796 kg ha⁻¹. It was significantly different from variety PSL2 with an average yield of 1564 kg ha⁻¹ and from variety PTT1 with an average yield of 1524 kg ha⁻¹. The harvest index (HI) for CNT1, PSL2 and PTT1 were 0.27, 0.23 and 0.20, respectively (Table 5).

There was also an interaction between planting date and variety ($P < 0.01$) for flooded rice yield production during the DS (Table 3). The first three ranks of interactions were PD1 × PTT1, PD1 × CNT1, and PD1 × PSL2 with an average yield of 3661, 3446, and 3217 kg ha⁻¹, respectively (Table 6). This indicated that the early planting PD1 and PD2 resulted in a higher yield than the late planting PD3 and PD4 (Table 6). Planting of variety PTT1 on November 9 resulted in the highest yield, but it was not significantly different from the variety CNT1 planted on the same date. Planting of flooded rice in the dry season should be done as soon as possible after the receding of flooded water to ensure that there is a sufficient water supply for the rice crop throughout the growing season.

Table 5

Effect of variety on the early rainy season and dry season crop in the deepwater area of Prachin Buri in Thailand, 2009–2010.

Variety	Yield (kg ha ⁻¹)	Biomass (kg ha ⁻¹)			Harvest index
		Above ground	Stem	Leaf	
Early rainy season crop					
Pitsanulok 2	3898a	8004c	2520c	1354b	0.49a
Pathum Thani 1	3759a	9022a	3136b	1573a	0.42b
Chai Nat 1	3529b	8509b	3455a	1635a	0.42b
Dry season crop					
Chai Nat1	1796a	5625a	2456ab	1233a	0.27a
Pitsanulok 2	1564b	5378a	2304b	1314a	0.23b
Pathum Thani 1	1524b	5477a	2590a	1222a	0.20c

Note: Means with a column followed by the same letter are not significantly different according to LSD test at $P < 0.05$.

4. Discussion

4.1. A comparison between early rainy season (ERS) and dry season (DS)

Average leaf, stem and above ground biomass for the ERS crop was higher than for the DS crop, especially for the later planting dates, due to insufficient water. In addition, there was an issue of saline water, due to an influx of sea water into the region as a result of the drought. Research has shown that salinity could injure rice growth at a high salt concentration (De Datta, 1981; Fageria, 1992; Jaiwal et al., 1997). Therefore, the overall average biomass for the ERS crop was higher than the biomass of the DS crop. The average above ground biomass at final harvest for the varieties PSL2, CNT1, and PTT1 for the ERS crop was 8004 (± 373), 8509 (± 218), and 9022 (± 597) kg ha⁻¹, respectively while the average above ground biomass for the DS crop over the four planting dates was 5378 (± 2473) kg ha⁻¹ for PSL2; 5625 (± 2395) kg ha⁻¹ for CNT1; and 5477 (± 2880) kg ha⁻¹ for PTT1. The standard deviation of rice biomass and yield for the DS crop was higher for the ERS crop due to a reduction of biomass of the late planting date caused by salinity and insufficient water supply for the late planting date of the dry season crop. This was the result of difference of water management during the ERS and the DS crop as explained previously.

4.2. Temperature and day length

The experiments were located at 13° 52' N, and 101° 09' E. The day lengths during the early rainy and dry season were not much different and ranged from 11 h 41 min to 12 h 34 min for the ERS crop and from 11 h 32 min to 12 h 15 min for the DS crop. The natural day length or photoperiod affecting rice development consists of the length of day light and the duration of civil twilight (De

Datta, 1981). For this location, the day length is longest on June 21 and shortest on December 21, with 12 h 50 min and 11 h 10 min, respectively (Ahrens, 2008). The optimum temperature for rice is 27 °C (George, 2007). The average maximum and minimum air temperatures were 33.9 °C (± 1.7) and 25.3 °C (± 0.9), respectively, for the early rainy season and 34.9 °C (± 2.5) and 24.0 °C (± 2.3), respectively for the dry season. There was, therefore, no significant difference in air temperature between the early rainy season and dry season production period. However, it should be noted that the maximum air temperatures were significantly higher than the optimum temperature of 25–30 °C for rice (Matthews, 1995; Paul, 1994; Singh, 2009; Sparks, 2009).

4.3. Planting date

The range of potential planting dates for flooded rice production in the deepwater area in Prachin Buri, Thailand was limited due to the flooding period during the rainy season and the limited water supply during the dry season. There was not much impact of planting date during the ERS on growth and yield. However, there is a risk of flooding or too much water during the harvest in October for the later planting dates (Catling, 1992). It is, therefore, recommended to plant as early as possible when the water is sufficient for land preparation and for rice growth during the seedling stage.

The biomass and yield of flooded rice during the DS depended on the available water supply throughout the growing season. It is, therefore, more important to select an appropriate planting date for the DS crop than for the ERS crop. In addition, an increase in the production of flooded rice also means a need for additional water for irrigation. Planting of flooded rice in this region should be done as early as possible for both the early rainy season and the dry season production.

Table 6

Effect of interaction between planting date (PD) and varieties (V) on flooded rice production in the deepwater area of Prachin Buri, Thailand in 2010.

Rank	Yield		Above ground biomass ^a		Harvest index	
	PD × V	Mean ^b (kg ha ⁻¹)	PD × V	Mean ^b (kg ha ⁻¹)	PD × V	Mean ^b
1	PD1 × PTT1	3661a	PD1 × PTT1	8898a	PD1 × CNT1	0.43a
2	PD1 × CNT1	3446ab	PD1 × CNT1	8001b	PD1 × PTT1	0.41ab
3	PD1 × PSL2	3217b	PD1 × PSL2	7951bc	PD1 × PSL2	0.41ab
4	PD2 × CNT1	2805c	PD2 × CNT1	6970cd	PD2 × CNT1	0.40ab
5	PD2 × PSL2	2557cd	PD2 × PSL2	6907d	PD2 × PSL2	0.37b
6	PD2 × PTT1	2246d	PD2 × PTT1	6803d	PD2 × PTT1	0.33c
7	PD3 × CNT1	884e	PD3 × CNT1	4949e	PD3 × CNT1	0.17d
8	PD3 × PSL2	397f	PD3 × PSL2	4030f	PD3 × PSL2	0.10e
9	PD3 × PTT1	113f	PD3 × PTT1	3371fg	PD3 × PTT1	0.04f
10	PD4 × PSL2	82f	PD4 × CNT1	2835g	PD4 × PSL2	0.03f
11	PD4 × PTT1	75f	PD4 × PSL2	2625g	PD4 × PTT1	0.02f
12	PD4 × CNT1	49f	PD4 × PTT1	2578g	PD4 × CNT1	0.02f

^a Above ground biomass includes leaf biomass, stem biomass and grain yield.^b Means with a column followed by the same letter are not significantly different according to LSD test at $P < 0.05$.

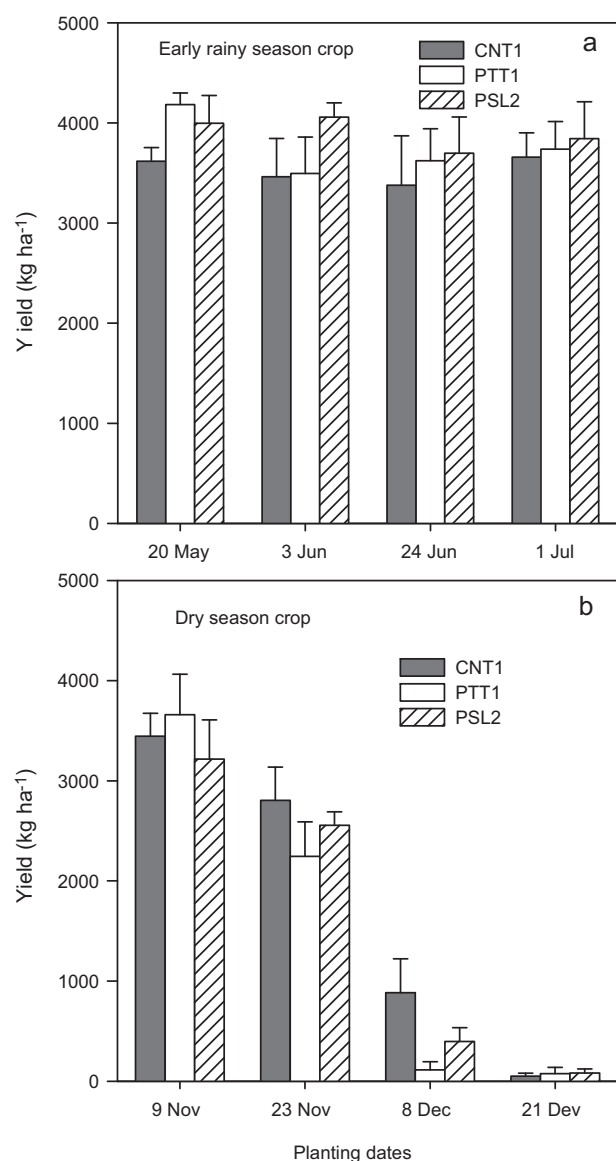


Fig. 2. Effect of variety and planting date on yield under early rainy season (a) and dry season (b) flooded rice production in the deepwater area of Prachin Buri, Thailand, 2009–2010.

4.4. Variety

There are currently more than 500 recommended modern rice varieties for flooded rice production across the world (Qi et al., 2009). Since 1959, only 29 modern rice varieties have been recommended for flooded rice production in Thailand. The growth duration from germination to harvest ranges from 90 to 140 days (Department of Agriculture, 2002). Most of these varieties are no longer planted in farmers' fields. Three widely adopted varieties for flooded rice production were used in this study, including the varieties CNT1, PTT1 and PSL2. They have similar characteristics with respect to growth duration and yield; but there were both advantages and disadvantages among those varieties. For example, PSL2 is suitable for early rainy season production because it is resistant to lodging, but CNT1 and PTT1 is better than PSL2 in term of cold tolerance. This information is based on observations and discussions with farmers. However, due to the constraints of the flooding period during the rainy season and water supply during the dry season, varieties that have shorter growth duration with a higher growth rate are preferred for planting in this area.

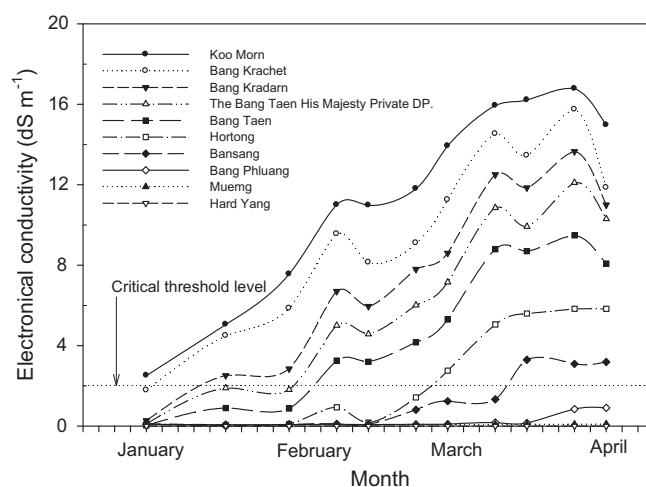


Fig. 3. Salinity of water at 10 stations along Bang Pakong River in the study area, Prachin Buri, Thailand, 2010.

There are other recommended early maturing rice varieties, but they could not compete with these three varieties in terms of yield (Department of Agriculture, 2002, 2004a,b).

4.5. Salinity

The demand for water for irrigation has increased with the expansion of flooded rice production. However, the water supply for flooded rice production is limited, especially during the dry season. The main channel that provides the irrigation water in the study area flows directly to the Gulf of Thailand. There is, therefore, a close relation between fresh water in the main channel and sea water in the Gulf of Thailand. The water supply for flooded rice production during the early rainy season is sufficient, due to the upstream water flow through the area to the gulf. However, during the dry season, the supply of water from upstream is less than the water used for irrigation. As a result, due to the increased pumpage for irrigation, water that is not replaced by the inflow from upstream will be replaced with seawater, causing an increase in salinity in the stream. Several salinity monitoring stations have been installed for measuring the salinity of Bang Pakong river along the study area. This includes the Bang Taen His Majesty Private Development Project, which was the location of this field experiment. Salinity could be detected as early as January at the downstream station and salinity continued to increase until April 7, 2010 (Fig. 3). Most crops, including rice, are sensitive to salinity. The critical salinity threshold for rice cultivation is 2.0 dS m^{-1} (De Datta, 1981; Fageria, 1992; Jaiwal et al., 1997). The salinity at the experimental site was first detected in late January. The salinity level of the water was higher than the critical level for rice as early as mid February and continued to increase until the highest level was reached in early April. However, there was no detection that the salinity levels exceeded the threshold level at the upper stream stations. This means that the potential planting date in the upper stream has a wider range than for the downstream area.

5. Conclusion

The results from this study indicate that during the transition process of deepwater rice to flooded rice production in Thailand, rice yield is affected by both planting date and variety. For the early rainy season production, yield is affected by variety. The highest yield over four planting dates was obtained with the variety PSL2 with an average yield of 3898 kg ha^{-1} . However, this was not significantly different from that variety PTT1 that had an average yield

of 3759 kg ha⁻¹. For flooded rice production during the dry season, the yield was affected by the interaction of planting date and variety. The highest yield was obtained for the variety PTT1 planted on November 9 with 3661 kg ha⁻¹. Overall, it can be concluded that the variety PSL2 is suitable for any planting date from May to July for flooded rice production during the early rainy season and that for the dry season the variety PTT1 should be planted in early November. Given the common problems for the transition from deep water rice production to flood rice production, it is expected that these recommendations are applicable to other regions that have similar water regimes and intrusion of sea water.

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